

**RESEARCH PROPOSAL TEMPLATE /**  
**THUYẾT MINH ĐỀ CƯƠNG NGHIÊN CỨU**

|                                             |             |
|---------------------------------------------|-------------|
| <b>Research title</b>                       |             |
| <b>Mentor</b>                               |             |
| <b>Members</b><br><i>(up to 3 students)</i> | Full name 1 |
|                                             | Full name 2 |
|                                             | Full name 3 |

**Abstract/Tóm tắt**

*Research objectives, research questions, research problems, research methodology (maximum 250 words) / Giới thiệu tóm tắt về đề tài (nêu mục tiêu nghiên cứu, câu hỏi nghiên cứu, vấn đề nghiên cứu và phương pháp nghiên cứu, không quá 250 từ).*

**Key words:**

**1. Introduction / Giới thiệu**

**1.1. Literature review / Tình hình nghiên cứu trong và ngoài nước**

*Review and discuss the related research work(s) done by Vietnamese and foreign scientists. / Nêu rõ những nghiên cứu do các nhà khoa học nước ngoài, trong nước tiến hành theo hướng nghiên cứu của đề tài và các kết quả đạt được, đồng thời nhận xét về những kết quả đã nêu.*

*Review the applications of related works in reality / Nêu rõ ứng dụng của các nghiên cứu này trên thực tế.*

**1.2. The limitation of current works / Những hạn chế của các nghiên cứu hiện tại**

*Clearly explain the limitations of current work and the necessity of your research. This helps establish the significance and originality of your study.*

*Nêu rõ những hạn chế của các nghiên cứu hiện tại để từ đó nêu lên được sự cần thiết của nghiên cứu này.*

**1.3. The necessity of the research / Sự cần thiết tiến hành nghiên cứu**

*Clearly define the problem or hypothesis to be addressed. / Nêu rõ vấn đề mà đề tài tập trung giải quyết.*

*Originality, relevance, and scientific significance of the question under investigation. / Phân tích tính mới, tính thời sự, ý nghĩa khoa học và sự cần thiết của vấn đề cần nghiên cứu.*

## **2. Research objectives / Mục tiêu của đề tài**

*Clearly outline the research objectives as a basis for determining research content and implementation plan. / Nêu rõ mục tiêu cần đạt được của đề tài, làm cơ sở xác định nội dung nghiên cứu và kế hoạch triển khai.*

## **3. Research scope / Phạm vi nghiên cứu**

*Clearly define the key study topics and the research scope. / Nêu rõ những nội dung nghiên cứu chính, phạm vi bao quát của đề tài.*

## **4. Feasibility of research / Tính khả thi của đề tài**

*Analyze the feasibility of research / Phân tích tính khả thi của đề tài.*

## **5. Approach and Method / Cách tiếp cận và phương pháp nghiên cứu**

*Describe research methodology and techniques appropriate for each issue of the research (including scheme/architecture/system models). / Mô tả chi tiết cách tiếp cận, phương pháp nghiên cứu, kỹ thuật sử dụng phù hợp với từng nội dung của đề tài nhằm đạt được mục tiêu của đề tài (bao gồm kiến trúc mô hình hoặc mô hình hệ thống).*

## **6. Research plan / Kế hoạch thực hiện nghiên cứu**

*Describe the research plan (Give a concise description of each task, and their expected results, schedule, and person in charge as in the following table) / Mô tả kế hoạch thực hiện của toàn bộ dự án nghiên cứu.*

| No. | Date                       | Task | Output | Person in charge |
|-----|----------------------------|------|--------|------------------|
| 1   | dd/mm/yyyy –<br>dd/mm/yyyy |      |        |                  |
| 2   |                            |      |        |                  |
| ... |                            |      |        |                  |

## **Computational Resource Requirements (if any)**

## **7. Expected results / Dự kiến kết quả đề tài**

*For example: theoretical, methodological, and practical implications; and plans to apply this research in practical. / Nêu dự kiến kết quả nghiên cứu (Đề xuất về mặt lý thuyết, phương pháp và ứng dụng, và khả năng ứng dụng vào thực tế, ...);*

## **References**

# **SAMPLE**

## **Abstract**

This research proposes an automated healthcare information system integrating clinical workflow management, billing and insurance claims processing, and fraud/anomaly detection with explainable and audit-ready alerts. The system supports an end-to-end healthcare process from patient registration and appointment scheduling to diagnosis, billing, and insurance claims. A key contribution of this study is the design of a two-layer Explainable Alert mechanism: (i) evidence-based explanations derived from immutable audit trails capturing user actions and data modifications, and (ii) model-based explanations generated from machine learning models using feature importance and local explainability techniques.

The research addresses a critical gap in existing healthcare fraud detection systems, which often provide risk scores without sufficient audit evidence to support real-world investigation and accountability. By integrating audit-aware features into machine learning models, the proposed system enhances both fraud detection accuracy and explainability. The methodology combines system design, database modeling, machine learning using Weka, and experimental evaluation on synthetic healthcare datasets. The expected outcome is a practical, auditable, and explainable healthcare fraud detection framework suitable for real-world deployment and academic dissemination.

**Keywords:** Healthcare Information System, Fraud Detection, Explainable AI, Audit Trail, Insurance Claims

## **1. Introduction / Giới thiệu**

### **1.1. Literature review**

International studies on healthcare fraud detection have extensively applied machine learning techniques such as logistic regression, decision trees, random forests, and support vector machines to insurance claim datasets. These studies demonstrate promising performance in identifying anomalous billing behaviors but often focus primarily on predictive accuracy rather than practical deployment.

Recent research emphasizes Explainable AI (XAI) methods, including LIME and SHAP, to improve transparency of model predictions. However, these explanations typically rely solely on statistical feature contributions and lack operational audit evidence required in healthcare and insurance investigations.

In Vietnam, research has largely focused on healthcare information systems (HIS), electronic medical records, and hospital management software, with limited emphasis on AI-driven fraud detection and explainability. Applications of existing works include hospital administration, insurance claim processing, and clinical decision support systems. Nevertheless, integration between AI detection and audit mechanisms remains minimal.

### **1.2. The limitation of current works**

Current fraud detection systems suffer from several limitations. First, many systems provide only risk scores without traceable evidence, making audit and dispute resolution difficult. Second, existing XAI approaches explain model behavior but do not capture user actions, data modifications, or workflow provenance. Finally, the separation between operational systems and AI models limits practical applicability in real healthcare environments. These limitations highlight the necessity of a unified, audit-aware, and explainable healthcare system.

### **1.3. The necessity of the research**

This research focuses on addressing the lack of audit-ready explainability in healthcare fraud detection systems. The originality lies in combining audit trail analysis with machine learning explainability to produce actionable and trustworthy alerts. The study is timely due to increasing digitalization of healthcare services and rising concerns over insurance fraud. Scientifically, the research contributes a novel framework that bridges operational audit requirements with AI-based anomaly detection.

### **Research objectives**

Design and implement an automated healthcare information system supporting end-to-end clinical and billing workflows.

- Develop an audit-aware fraud and anomaly detection mechanism for billing and insurance claims.
- Propose a two-layer explainable alert model combining audit evidence and machine learning explanations.
- Evaluate the effectiveness of audit-aware features in improving fraud detection performance and explainability.

### **Research scope**

The study focuses on a mini healthcare system for clinics or hospitals, covering patient management, appointments, encounters, billing, and insurance claims. Machine learning models are applied to claim and billing data using synthetic datasets. The research does not aim to deploy a full-scale national HIS but rather a feasible and extensible prototype.

### **Approach and Method**

The research adopts a system-oriented and experimental approach. First, a healthcare workflow system is designed using Java MVC architecture. Second, an immutable audit trail mechanism is implemented at the database level. Third, machine learning models are trained using Weka to detect fraudulent or anomalous claims. Finally, explainable alerts are generated by combining audit evidence with model-based explanations. Experimental evaluation is conducted using standard classification metrics and case studies.

### **Research plan**

| No. | Date   | Task                            | Output                        | Person in charge |
|-----|--------|---------------------------------|-------------------------------|------------------|
| 1   | Week 1 | System design & database schema | System architecture, ERD      | Research team    |
| 2   | Week 1 | Core modules & audit trail      | Working HIS prototype         | Research team    |
| 3   | Week 2 | ML model training & integration | Fraud detection module        | Research team    |
| 4   | Week 2 | Explainable alert & evaluation  | Experimental results & report | Research team    |

### Computational Resource Requirements

A standard development computer with Java runtime environment, SQL Server, and sufficient memory to train machine learning models on synthetic datasets is required. No specialized hardware is necessary.

### Expected results

The research is expected to deliver a functional healthcare information system with integrated audit-aware fraud detection. Theoretical contributions include a novel explainable alert framework, while practical outcomes include a deployable prototype and experimental evidence of improved explainability and auditability. The results can be applied in real healthcare and insurance environments and serve as a foundation for further academic research.

### References

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